Swirl Diffuser SWR



Description
Swirl jet diffuser with radial fixed blades **SWR**.

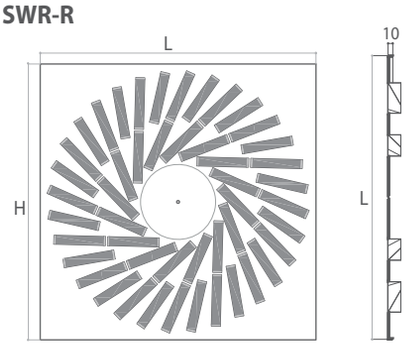
Application
Supply/ exhaust of the air flow

Characteristics
Recommended for rooms between 2.60 to 4.00 meters in height.
Depending on the technical and architectural requirements, the diffuser is produced with round or rectangular front **R/C** .
Blade design, radial layout and swirl action, result in a high level induction rate.
The diffuser should be embedded into the ceiling to ensure a stable horizontal discharge of air.
The diffuser is supplied with seals, screw fastening and masking plastic lid.
Blades are available in two colors: white RAL 9016 and black RAL9005 .

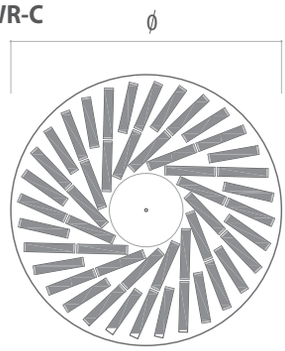
Accessories
Insulated/Non-insulated adapter **ADP IZ/N.**

Materials and finishing
Pickled steel sheet electrostatic field painted white RAL 9016.
Other RAL colors available on request.

Technical design
SWR-R



SWR-C



Assembly Method
Paneled ceiling 600 x 600 mm mounting, fitted with a central screw M6 x 100 mm

Swirl Diffuser SWR

Selection table SWR

SWR - 8 slots

Air flow (m^3/h)	Ak [m^2]	300 x 300 / D300	
		0.007	
		Exhaust	Supply
100	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.0 <20 9.0	4.0 <20 10.0
125	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.0 <20 14.0	5.0 <20 12.0
150	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.0 22.0 18.0	6.0 <20 14.0
175	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.9 28.0 28.0	6.9 <20 16.0
200	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.9 30.0 30.0	7.9 22.0 17.0
250	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.9 38.0 52.0	9.9 26.0 22.0
300	Veff[m/s] NR [dB(A)] ΔPt [pa]		11.9 35.0 35.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]		15.9 42.0 60.0

SWR - 16 slots

Air flow (m^3/h)	Ak [m^2]	400 x 400/D400	
		0.014	
		Exhaust	Supply
200	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.0 15.0 10.0	4.0 <20 5.0
250	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.0 25.0 15.0	5.0 <20 6.0
300	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.0 30.0 25.0	6.0 <20.0 10.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.9 38.0 40.0	7.9 26.0 15.0
500	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.9 46.0 60.0	9.9 33.0 25.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	11.9 48.0 75.0	11.9 40.0 36.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]		13.9 45.0 50.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]		15.9 50.0 65.0

SWR - 24 slots

Air flow (m^3/h)	Ak [m^2]	500 x 500 / D500	
		0.021	
		Exhaust	Supply
200	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.6 <25 7.0	2.6 <20 3.0
300	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.0 28.0 13.0	4.0 <20 4.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.3 35.0 25.0	5.3 <20 7.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.9 45.0 52.0	7.9 30.0 17.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.3 50.0 70.0	9.3 33.0 20.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	10.6 55.0 80.0	10.6 42.0 32.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]		13.2 50.0 52.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]		15.9 60.0 65.0

SWR - 24 slots

Air flow (m^3/h)	Ak [m^2]	595 x 595 / D595	
		0.0295	
		Exhaust	Supply
250	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.4 <20 4.0	2.4 <20 3.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.8 24.0 12.0	3.8 <20 4.0
500	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.7 30.0 15.0	4.7 <20 6.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.6 33.0 20.0	5.6 <20 8.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.6 36.0 25.0	6.6 23.0 9.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.5 43.0 35.0	7.5 30.0 15.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]		9.4 35.0 20.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]		11.3 40.0 30.0

Selection table SWR

SWR - 48 slots

Air flow (m^3/h)	Ak [m^2]	595 x 595 / D595	
		0.0390	
		Exhaust	Supply
350	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.5 <20 6.0	2.5 <20 3.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	2.8 23.0 12.0	2.8 <20 4.0
500	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.6 27.0 14.0	3.6 <20 5.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.3 30.0 17.0	4.3 <20 6.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.0 33.0 20.0	5.0 23.0 7.5
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.7 36.0 28.0	5.7 25.0 8.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.1 43.0 45.0	7.1 35.0 15.0
1400	Veff[m/s] NR [dB(A)] ΔPt [pa]		10.0 45.0 30.0

SWR - 24 slots

Air flow (m^3/h)	Ak [m^2]	623 x 623 / D623	
		0.0295	
		Exhaust	Supply
300	Veff[m/s] NR [dB(A)] ΔPt [pa]		2.8 <20 6.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.8 <20 4.0	3.8 27.0 12.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.6 <20 8.0	5.6 33.0 18.0
700	Veff[m/s] NR [dB(A)] ΔPt [pa]	6.6 25.0 10.0	6.6 35.0 23.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.5 30.0 15.0	7.5 42.0 33.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]	9.4 38.0 28.0	9.4 46.0 45.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]	11.3 42.0 32.0	11.3 53.0 70.0
1400	Veff[m/s] NR [dB(A)] ΔPt [pa]	13.2 50.0 40.0	

SWR - 54 slots

Air flow (m^3/h)	Ak [m^2]	623 x 623 / D623	
		0.047	
		Exhaust	Supply
300	Veff[m/s] NR [dB(A)] ΔPt [pa]		1.8 <20 3.0
400	Veff[m/s] NR [dB(A)] ΔPt [pa]		2.4 <20 6.0
600	Veff[m/s] NR [dB(A)] ΔPt [pa]	3.5 <20 3.0	3.5 27.0 13.0
800	Veff[m/s] NR [dB(A)] ΔPt [pa]	4.7 25.0 6.0	4.7 35.0 23.0
1000	Veff[m/s] NR [dB(A)] ΔPt [pa]	5.9 30.0 8.5	5.9 43.0 40.0
1200	Veff[m/s] NR [dB(A)] ΔPt [pa]	7.1 35.0 12.0	7.1 46.0 50.0
1500	Veff[m/s] NR [dB(A)] ΔPt [pa]	8.9 42.0 20.0	8.9 52.0 73.0
1800	Veff[m/s] NR [dB(A)] ΔPt [pa]	10.6 50.0 30.0	

Annotation

- Ak [m²] - Free surface
- Veff[m/s] - Air velocity inside the diffuser
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss

Order codes

Type	-	R/C	+	ADP IZ/N	/	Dimensions - No. of slots	/	Blades color	/	O
SWR		Rectangular / Circular		Insulated/Non-insulated adapter		300 x 300 / D300 - 8 400 x 400 / D400 - 16 500 x 500 / D500 - 24 595 x 595 / D595 - 24 595 x 595 / D595 - 48 623 x 623 / D623 - 24 623 x 623 / D623 - 54		White RAL9016 Black RAL9005		**0** Painted white RAL 9016 **R** Other RAL paints